OCR Benchmark Report

PaddleOCR PP-OCRv5 Multilingual vs Tesseract 5

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Executive Summary

Using the supplied OCR output files, PaddleOCR PP-OCRv5 Multilingual produced the stronger measured result across CER, WER, key-field recall, Turkish-character preservation, table/form marker recall, and reading-order similarity. Tesseract 5 remains easier to deploy and is CPU-only, but its raw output loses more Turkish diacritics and introduces more layout noise on these municipal forms.

Metric	PaddleOCR PP-OCRv5	Tesseract 5	Better
Character Error Rate	0.271	0.356	Lower is better
Word Error Rate	0.268	0.644	Lower is better
Key-field accuracy	72.0%	50.0%	Higher is better
Turkish-character accuracy	91.9%	58.5%	Higher is better
Table marker preservation	75.7%	42.9%	Higher is better
Reading-order preservation	72.4%	49.6%	Higher is better

Dataset And Ground Truth

The benchmark folder contains 27 files across 42 pages (21 PDFs and 6 standalone images). CER/WER were computed only for 17 PDFs with extractable embedded text of at least 250 characters. Scanned PDFs and image-only files were included in key-field/table checks but excluded from CER/WER because no manual ground truth was supplied.

All text-quality scores use the provided output files as-is. No OCR output was corrected before scoring.

Detailed Measurement Definitions

- Character Error Rate: Levenshtein character distance divided by reference character count after Unicode normalization, case folding, and whitespace collapse.
- Word Error Rate: Levenshtein word distance divided by reference word count on Turkish-aware word tokens.
- Key-field accuracy: recall of curated document labels and key field names across the supplied benchmark files, matched accent-insensitively to avoid double-penalizing Turkish marks.
- Turkish-character accuracy: recall of expected Turkish-specific characters in the embedded-text reference subset, measured from the raw OCR output.
- Table preservation: recall of key table/form labels on table-heavy documents.
- Reading-order preservation: word-sequence similarity against embedded-text references after accent-insensitive folding.

Model-Level Results

Area	PaddleOCR PP-OCRv5 Multilingual	Tesseract 5
Observed strength	Best measured raw text quality in this artifact set; strong key-field recall and much better Turkish-character preservation than Tesseract.	Simple CPU deployment; native CLI is easy to call from scripts and does not require GPU memory.
Observed weakness	Heavier Python/Paddle dependency stack; some Turkish letters are still substituted with non-Turkish accented Latin characters; runtime footprint not captured.	Plain text output contains layout noise, weak table structure, and many Turkish diacritics are lost or substituted.
Average processing time per page	Not captured in supplied output/log files.	Not captured in supplied output/log files.
Peak RAM	Not captured in supplied output/log files.	Not captured in supplied output/log files.
Peak VRAM	Not captured. PaddleOCR can run CPU or GPU depending on PaddlePaddle runtime.	0 VRAM expected for standard Tesseract CLI/Python use; CPU-only OCR engine.
CPU/GPU compatibility	Python/PaddlePaddle stack; CPU and CUDA GPU deployments are possible, but dependency setup is heavier.	CPU-first native binary. GPU is not required and normally not used.
Output format	Script writes plain text; PaddleOCR result objects can also expose structured recognition fields.	Script writes plain text; Tesseract can also emit hOCR, TSV, searchable PDF, and box formats when configured.
Integration difficulty	Medium-high: Python package, model downloads, Paddle runtime, optional GPU/CUDA alignment.	Low-medium: install native Tesseract plus language data and call CLI or wrapper.
License suitability	Suitable for commercial/internal use under Apache-2.0, subject to preserving notices and checking bundled model/data terms.	Suitable for commercial/internal use under Apache-2.0, subject to preserving notices and traineddata terms.

Per-Document Text Metrics

The following rows show the CER/WER subset only: PDFs where embedded text was available. Lower CER/WER is better.

File	Paddle CER	Tess CER	Paddle WER	Tess WER	Ref chars
company_registration_documents.pdf	0.085	0.204	0.111	0.465	1535
Fire_safety_reports.pdf	0.844	0.878	0.857	1.259	3881
Fire_safety_reports_2.pdf	0.131	0.199	0.154	0.551	582
Fire_safety_reports_3.pdf	0.080	0.136	0.098	0.462	1046
Lease_agreements.pdf	0.015	0.104	0.038	0.479	8612
Lease_agreements_2.pdf	0.202	0.321	0.319	0.664	736
Licence_application_forms.pdf	0.323	0.570	0.045	0.660	2318
Occupancy_permits_2.pdf	0.483	0.568	0.546	0.777	1950
Occupancy_permits_3.pdf	0.274	0.312	0.254	0.676	1174
Occupancy_permits_4.pdf	0.260	0.306	0.335	0.731	1476
petition_1.pdf	0.497	0.573	0.308	0.785	578
petition_2.pdf	0.454	0.467	0.089	1.200	540
Table_based_forms_1.pdf	0.698	0.691	0.897	0.918	1115
Table_based_forms_2.pdf	0.337	0.441	0.394	0.725	1165
Table_based_forms_3.pdf	0.082	0.166	0.135	0.499	4680
Tax_certificates.pdf	0.426	0.503	0.651	0.738	632
Tax_certificates_2.pdf	0.365	0.479	0.606	0.810	764

Key-Field Accuracy By Document

File	Paddle	Tesseract	Fields
company_registration_documents.pdf	5/8 (62.5%)	3/8 (37.5%)	text/form
Fire_safety_reports.pdf	7/7 (100.0%)	7/7 (100.0%)	table/form
Fire_safety_reports_2.pdf	6/6 (100.0%)	5/6 (83.3%)	text/form
Fire_safety_reports_3.pdf	6/6 (100.0%)	4/6 (66.7%)	text/form
Lease_agreements.pdf	6/7 (85.7%)	5/7 (71.4%)	table/form
Lease_agreements_2.pdf	4/6 (66.7%)	4/6 (66.7%)	table/form
Licence_application_forms.pdf	3/6 (50.0%)	3/6 (50.0%)	text/form
Occupancy_permits_2.pdf	3/5 (60.0%)	2/5 (40.0%)	text/form
Occupancy_permits_3.pdf	4/5 (80.0%)	2/5 (40.0%)	text/form
Occupancy_permits_4.pdf	1/5 (20.0%)	1/5 (20.0%)	text/form
petition_1.pdf	5/5 (100.0%)	5/5 (100.0%)	text/form
petition_2.pdf	4/5 (80.0%)	5/5 (100.0%)	text/form
Table_based_forms_1.pdf	6/6 (100.0%)	2/6 (33.3%)	table/form
Table_based_forms_2.pdf	4/5 (80.0%)	0/5 (0.0%)	table/form
Table_based_forms_3.pdf	3/5 (60.0%)	4/5 (80.0%)	table/form
Tax_certificates.pdf	4/5 (80.0%)	2/5 (40.0%)	table/form
Tax_certificates_2.pdf	4/5 (80.0%)	3/5 (60.0%)	table/form
ashissedevri-300x215.jpg	3/5 (60.0%)	0/5 (0.0%)	table/form
İskan_ve_Yapı_Ruhsat(2000_yılı)_1_75629.jpeg	5/7 (71.4%)	2/7 (28.6%)	table/form
İşyeri Ruhsat İntibak Dilekçesi.png	5/6 (83.3%)	4/6 (66.7%)	text/form
vergi-levhasi.jpg	0/5 (0.0%)	0/5 (0.0%)	table/form
yangin_raporu.png	7/7 (100.0%)	1/7 (14.3%)	table/form
yapı_denetim_izin.jpg	0/5 (0.0%)	2/5 (40.0%)	text/form

Interpretation

- PaddleOCR PP-OCRv5 is the stronger candidate for this dataset if accuracy is the priority.
- Tesseract 5 is still attractive when installation simplicity, CPU-only operation, and predictable licensing/deployment matter more than accuracy.
- Neither provided output captures peak memory or processing time. Add process-level RAM/GPU sampling to both scripts before using this benchmark for production cost planning.
- For table-heavy municipal forms, neither plain-text output preserves true table geometry. If table extraction is a requirement, use structured OCR outputs with bounding boxes, hOCR/TSV, or a layout model.

Runtime And Memory Instrumentation Gap

Average processing time per page, peak RAM, and peak VRAM could not be measured from the two supplied OCR text files because they contain extracted text only, not execution logs. A defensible runtime benchmark should rerun each model with: per-page start/end timestamps, process RSS sampling, and GPU memory sampling via nvidia-smi for GPU-backed PaddleOCR.

Local check	Value
Tesseract executable	tesseract v5.5.0.20241111
GPU visible locally	NVIDIA GeForce GTX 1660 Ti, 6144 MiB
PaddleOCR runtime	Not installed in the local workspace Python checked during this report; provided output file was used instead.

Sources And License Notes

- PaddleOCR repository and license: https://github.com/PaddlePaddle/PaddleOCR and https://github.com/PaddlePaddle/PaddleOCR/blob/main/LICENSE
- Tesseract repository and license: https://github.com/tesseract-ocr/tesseract and https://github.com/tesseract-ocr/tesseract/blob/main/LICENSE
- Local PaddleOCR output: C:\Users\asus\OneDrive\Desktop\mock_agent_proj_trainig\ocr_models\output\output_ocr\paddleocr_pp_ocrv5_multilingual\paddleocr_pp_ocrv5_multilingual_output.txt
- Local Tesseract output: C:\Users\asus\OneDrive\Desktop\mock_agent_proj_trainig\ocr_models\output\output_ocr\tesseract_ocr\tesseract_ocr_output.txt
- Benchmark documents: C:\Users\asus\OneDrive\Desktop\mock_agent_proj_trainig\ocr_models\benchmark_mock_docs

Appendix: Limitations

- CER/WER use embedded PDF text as reference where available. No manual ground truth was supplied for scanned PDFs or standalone images.
- Runtime, peak RAM, and peak VRAM were not present in the provided OCR output files, so those metrics are reported as not captured.
- PaddleOCR output is evaluated exactly as supplied. It preserves many Turkish characters, but some substitutions such as non-Turkish accented Latin letters remain visible.